Dynamic Pricing and Demand Forecasting for Sports Facility Aggregators

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***Abstract*—Sports facility platforms need booking forecasts and evidence about customer responses before they change prices. We evaluate a forecasting and constrained-pricing pipeline on 153,300 synthetic hourly slots across 14 venues and 730 days. The final 146 days are reserved for testing. Chronological validation compares logistic regression, Light Gradient Boosting Machine (LightGBM), a long short-term memory network (LSTM), historical booking rates and a seasonal autoregressive integrated moving average model with exogenous regressors. A separate price-response model supports profit optimization within owner limits. LightGBM achieves test Brier score 0.1951, compared with 0.2107 for historical booking rates, and logarithmic loss 0.5747. However, paired tests do not establish an advantage over the LSTM, and its calibration-slope confidence interval excludes the target of one. It therefore fails the requirement to improve on every baseline and meet the calibration target. The estimated price response predicts revenue growth of 4.87%, with a 95% confidence interval of 3.28–6.79%. The known simulator mechanism gives 1.15% for the same prices, outside that interval. Estimated off-peak occupancy changes by -0.31 percentage points, with an interval that includes zero. Estimated cancellations decrease by 0.51 percentage points, but the joint requirement for higher off-peak occupancy and cancellation control is not met. The common evaluation record exposes errors that forecast scores alone miss. The results support further validation of the demand response before testing the policy with real customers.**

***Index Terms*—dynamic pricing; demand forecasting; sports facilities; synthetic data; booking probability; policy evaluation**

I. Introduction

Sports facility aggregators sell capacity that expires. An empty court or pitch cannot recover revenue from a slot after its start time. Demand changes with time, weather, facility type and price. Owners need booking forecasts to plan operations, but a pricing decision also requires an estimate of how customers respond to a price change. A model can predict bookings accurately while estimating that response poorly. Forecast accuracy and pricing performance therefore need separate tests on the same offered slots.

The distinction matters for managers and customers. A price increase can raise revenue per booking while reducing the number of bookings. A discount can attract demand, but its revenue effect depends on the size of the response. Cancellations can reduce the value of an apparently successful booking. Owners therefore need evidence for revenue, utilization and cancellations, with clear denominators and uncertainty estimates. A forecast score or an estimated revenue increase alone cannot describe all these outcomes.

Research on golf provides a close operational comparison. Kimes treats tee times as perishable capacity and explains how revenue management can apply when demand varies and customers can shift their purchase or playing time [1]. Kimes and Schruben study tee-time intervals through simulation [2]. They show why tighter scheduling does not necessarily increase revenue: variation in the pace of play can affect the capacity that a course can deliver. Courts and pitches have different scheduling rules, but the same general issue applies. A price rule must operate on a defined inventory rather than assume that all nominal capacity is interchangeable.

Pekgün, Uyar and Garner examine challenges in applying pricing and revenue management to golf [3]. Their discussion links demand forecasting, price optimization and tee-sheet construction. This work already establishes a connection between forecasting and pricing in recreational sport. Our research question is narrower: can a fitted booking model, an estimated price response and a constrained optimizer pass explicit checks on one common inventory? The focus is on checking the linked components, including cases where a favorable forecast score does not support a reliable policy estimate.

Sports-ticket research provides another relevant comparison. Shapiro and Drayer examine dynamic ticket prices and secondary-market prices in Major League Baseball [4]. Their later study examines determinants of prices in both markets [5]. These studies show that sports prices depend on event and market conditions. Their units are tickets for scheduled games, whereas facility aggregators repeatedly offer venue-time slots. This difference affects the inventory, the timing of a decision and the outcome being predicted. Ticket-pricing evidence motivates a sports application, but does not directly validate a booking-probability model for courts or pitches.

Recent system research also addresses the booking process. Nagamani et al. describe an online sports-slot reservation system with weather-based pricing and team matching [6]. Such a system connects reservations with contextual pricing, but a system description does not by itself test the accuracy of a counterfactual revenue estimate. Madden et al. forecast daily recreational-trail traffic using machine learning and weather information [7]. Their outcome is visitor volume rather than priced slot inventory. Together, these studies motivate contextual booking forecasts and a separate assessment of the demand response used by a price optimizer.

Marketplace research explains why owner control and cancellations belong in that assessment. Filippas et al. connect pricing authority with revenue, asset use and provider participation [8]. Li and Zheng study pricing under uncertain demand, external information and finite inventory [9]. Xu et al.

include cancellations in a dynamic pricing model [10]. Sports facilities share limited capacity and uncertain demand with these settings. Their supply, however, stays at a fixed venue. We retain owner-defined price limits and report cancellation outcomes without assuming that transport or product-market effects transfer to sports bookings.

Probability forecasts need scores that evaluate the probabilities used in later decisions. Brier introduced a score for probability forecasts [11], and Gneiting and Raftery explain the properties of proper scoring rules [12]. These sources support our use of Brier score and logarithmic loss. Calibration gives an additional check: slots assigned a probability near 0.60 should have a booking frequency near 0.60. Guo et al. distinguish calibration from other measures of predictive performance [13]. That distinction matters when an optimizer uses probabilities to compare expected revenues at alternative prices.

Validation must also follow the order in which information becomes available. Bergmeir and Benítez discuss evaluation for time-series predictors [14]. We therefore use chronological development folds and reserve the final period for testing. Hyndman and Koehler explain weaknesses in several forecast-error measures [15]. Because the booking target includes zeros, we use probability losses for individual slots and assess daily booking counts separately. The comparisons use common test observations so that a change in the sample cannot explain a difference between models.

Pricing requires further assumptions. Historical prices can reflect the same demand conditions that influence bookings. An association between high prices and many bookings can therefore differ from the effect of raising a price. Chernozhukov et al. develop orthogonalized estimation with cross-fitting to reduce bias from learned nuisance functions under stated assumptions [16]. We use this approach to estimate segment-level price responses. A simulation then permits an unusual check: the chosen prices can be evaluated both with the fitted response and with the generator's known booking mechanism. Their agreement is evidence about the policy estimate, separate from forecast accuracy.

The offered slot is the unit of analysis. A table that contains only completed bookings omits the empty capacity that an occupancy forecast must predict. It can also change the apparent relationship between prices and demand. We keep unbooked slots and record cancellations separately. This preserves the distinction between the probability of a booking and the probability that a booking is later cancelled. Policy comparisons then use the same offered inventory and a fixed denominator for each outcome. These choices make operational results comparable across models and prices.

The resulting research gap is a common evaluation of repeated sports slots that checks forecast quality, estimated revenue and operational outcomes together. Golf studies connect capacity and pricing [1]–[3]; ticket studies examine event markets [4], [5]; recreational forecasting and booking-system studies address other parts of the task [6], [7]. We assess the linked pipeline on one fixed inventory and retain unsuccessful checks. This provides a specific test of whether the estimated demand response supports the policy gains that the optimizer reports.

The aim is to evaluate a reproducible forecasting and constrained-pricing pipeline for sports slots, including its prediction errors, revenue estimates and operational limits.

1) RQ1: Does the validation-selected model improve booking-probability accuracy over every evaluated baseline, while meeting the stated skill and calibration criteria?
2) RQ2: What revenue change does the constrained pricing policy produce, and how does the estimate compare with the known simulator effect?
3) RQ3: Does the policy increase weekday off-peak occupancy without an unacceptable increase in cancellations?

H1 requires lower Brier score and log-loss than every baseline at adjusted $p < 0.05$, positive Brier skill, and a calibration-slope interval containing one. H2 requires the lower bound of the estimated revenue-change interval to exceed zero. Agreement with the known simulator effect is an additional check for RQ2. H3 requires a positive lower bound for off-peak occupancy change. It also requires the cancellation-change upper bound to stay below half the training weekly cancellation-rate standard deviation. Section III defines the calculations, and all intervals use 95% confidence.

The contribution is a reproducible evaluation record linking booking forecasts, segment-level price responses and constrained policy outcomes. It covers 153,300 offered slots at 14 synthetic venues, including 30,660 held-out test slots. Saved predictions support paired model comparisons, and saved response estimates document the inputs to each pricing decision. Evaluating the same chosen prices under the known generator reveals errors that remain hidden in forecast scores or estimated revenue alone. This controlled study supports a direct audit of the implemented pipeline; validation with real customer records remains a separate step.

Section II reviews pricing and forecasting studies. Section III describes the data, models, price-response estimates and evaluation protocol. Section IV reports the results and failed checks. Section V answers the research questions and states the implications and limitations. Sections VI and VII give data and code access details.

II. Literature review

The core review covers 13 research studies and 10 reviews published from 2021 to 2025. Its search used scholarly journals and publisher or institutional records from ScienceDirect, Springer Nature, INFORMS, Wiley, ACM, MDPI and related indexes. Conference papers, book chapters and unpublished preprints were outside that review corpus. The Introduction adds domain studies on golf, sports tickets and reservation systems, alongside methodological sources for the experiment. The review asks how forecasting and pricing evidence can inform a system for repeated sports-facility bookings.

A. Pricing and marketplace mechanisms

Hu, Hu and Zhu use a two-period game-theoretic model to study price responses on both sides of a ride-hailing platform

[17]. Riders and drivers respond at different speeds, and information sharing changes the possible surge-pricing patterns. Their model explains why the timing of a response matters. Its two-sided structure is useful, but mobile drivers differ from fixed sports venues. A sports platform must therefore test its own booking response over time.

Garg and Nazerzadeh study driver incentives with a dynamic stochastic model and marketplace data [18]. They show that additive surge payments can align driver incentives better than multiplicative payments. The analysis links prices to the time consumed by trips. Miao et al. use a quasi-experiment and causal-forest analysis after surge pricing starts in two Chinese cities [19]. Weekly driver revenue increases, but daily revenue decreases as competition grows and drivers work more days. These studies show that one favorable aggregate outcome can conceal other changes in behavior.

Jiao and Ramezani optimize discounts in an e-hailing model with differences in price sensitivity, service preferences and impatience [20]. Under their assumptions, dynamic discounts can increase shared rides and profit while reducing fleet requirements. Lei and Ukkusuri formulate ride-hailing prices as a sequential decision problem and develop scalable reinforcement-learning methods [21]. Their small and large experiments assess computational performance. Both studies support policies that respond to changing conditions. However, matching, mobility and model-training requirements limit direct transfer to a platform with fixed venues and sparse booking histories.

Xu, Saberi and Liu explicitly model cancellations before and after order confirmation [10]. They compare price and penalty rules and find higher profit and taxi use under some relaxed restrictions. Their conclusions depend on the behavioral assumptions and transport setting. Liang, Jiang and Lo use a dynamic equilibrium model with uncertain travel times and simultaneous departure-time and mode choices [22]. Their results connect uncertainty with demand and price decisions. For sports slots, both papers motivate explicit cancellation and uncertainty checks, with effects measured under the assumptions actually used.

Pricing authority can also change provider behavior. Filippas, Jagabathula and Sundararajan combine experimental and quasi-experimental evidence from a marketplace that changes its pricing system [8]. Centralized prices increase asset use and revenue, but providers also reduce availability or exit. Partial pricing control produces similar revenue gains with fewer provider reactions. The evidence concerns one platform, yet it gives a reason to preserve owner-defined limits in sports pricing. A revenue optimizer that ignores owner participation can omit an important operational response.

Li and Zheng study a seller with finite inventory, new external information and an unknown demand function [9]. Their analytical framework compares online decisions with a regime that knows demand, then develops pricing algorithms. It connects demand learning with capacity constraints, although its single-seller structure is more restricted than an aggregator. A sports slot is also a finite unit whose selling period ends. The study therefore supports an explicit separation between a demand estimate, the price rule and a benchmark with known information.

B. Sports demand and information feedback

Madden et al. predict daily foot traffic at three recreational trails with machine learning and weather variables such as rain and wind [7]. Their results support the use of environmental and contextual inputs for recreation demand. Daily visitor counts, however, differ from hourly offered slots, and the study does not optimize prices. Quansah, Buraimo and Lang use regression models for Premier League ticket prices across five seasons [23]. Opponents and local derbies affect prices, while team performance has an asymmetric effect on more expensive tickets. The sports setting is relevant, but event tickets do not represent repeated venue-time inventory.

Reviews and ratings introduce another source of price feedback. Shin, Vaccari and Zeevi model how customers learn product quality from quality-based and value-based reviews [24]. Their fluid approximation and marketplace-data simulation identify different pricing regimes. Carnehl, Stenzel and Schmidt examine how prices affect ratings and how ratings then affect demand [25]. Their analytical model includes customer selection and differences between consumer and seller objectives. Both studies explain a possible feedback mechanism, but their product-market assumptions need separate tests for sports facilities.

The current simulator does not include review dynamics or strategic provider entry. It focuses on time, venue attributes, weather, price and cancellations. This narrower scope makes the measured effects easier to trace to the specified mechanism. It also means that the policy cannot represent every response described in the marketplace literature. Future real-data work should record such omitted responses before adding them to a pricing model.

C. Forecasting and occupancy reviews

Petropoulos et al. review forecasting methods, evaluation, uncertainty and applications [26]. Their broad account supports a comparison of alternatives instead of an assumption that one algorithm works best everywhere. Benidis et al. review deep forecasting, including recurrent, convolutional, attention-based and probabilistic methods [27]. They also discuss data requirements, interpretation and model selection. Together, these reviews support classical baselines and a neural alternative, evaluated at the horizon and data scale relevant to the task.

Zhang, Wu and Calautit review 160 studies of building-occupancy prediction [28]. Their coverage links data collection, temporal and environmental variables, machine learning and validation. Li et al. review 263 occupancy studies and examine the interaction between data-collection technologies, algorithms and evaluation [29]. Hybrid methods and connected sensors receive attention, but reliable data and validation remain necessary. Building occupancy is not a booking record. The useful analogy is the need to predict presence or utilization from historical patterns and context, with clear limits on transfer between domains.

Chen et al. organize research on long-sequence forecasting and discuss benchmark resources and generalization across datasets and horizons [30]. Su et al. review Transformer-based long-term forecasting, including attention methods, computational efficiency and data limitations [31]. Song et al. cover deep time-series forecasting from 2014 to 2024, including temporal and cross-variable relationships [32]. These reviews support consistent horizons, common metrics and meaningful baselines. They also show why a four-step contextual sequence should not be interpreted as a test of long-range neural memory. The present pilot compares a short recurrent model and does not test a Transformer.

Muth, Lingenfelder and Nufer review 64 studies of demand prediction under macroeconomic volatility [33]. They discuss forward-looking information, changing demand conditions, combined models and neural methods. Douaioui et al. compare machine-learning and deep-learning approaches for supply-chain demand forecasting [34]. Their review emphasizes the relationship between the method, available data and operating constraints. These findings support context-specific evaluation for sports demand. A newer or more complex model still needs evidence of improvement under the relevant validation design.

Yang and Xia review 158 articles on pricing in the sharing economy [35]. They classify business models and pricing authority across accommodation, transport and other platforms. The reviewed methods include optimization, equilibrium models, auctions, regression and causal empirical designs. Information asymmetry, competition and empirical validation remain open issues. For sports aggregators, the review connects underused capacity with pricing and owner control, while leaving direct evidence for repeated sports slots limited.

D. Synthesis and research gap

The literature identifies several requirements for a sports-slot system. Forecasting should represent venue differences and relevant time, calendar and weather information [7], [26]–[29]. Price-response estimation must separate price effects from the demand conditions used to choose historical prices [9], [17], [19]. Cancellation models and owner limits are also necessary parts of the evaluation [8], [10], [35]. These requirements concern different components, so a good result in one component does not verify the complete pipeline.

Direct evidence for repeated bookable sports slots remains limited in this review. Ticket studies concern events [4], [5], [23], recreational-trail forecasting omits price optimization [7], and transport studies include mobile supply or matching [17], [20], [21]. We combine booking forecasts, an estimated price response, cancellation prediction and a constrained price optimizer on one synthetic inventory. RQ1 assesses forecasts, RQ2 assesses revenue calculations, and RQ3 assesses off-peak use and cancellations. The known simulator response permits a check of the estimated policy effect. Validation on real aggregator records remains a separate research stage.

III. Materials and Methods

A. Study design

The study uses a quantitative, two-stage design. Stage 1 estimates booking probability from slot characteristics. Stage 2 applies an estimated price-response function and chooses a price within stated constraints. This separates forecast evaluation from the counterfactual assumptions needed for pricing. Forecasting, price sensitivity and owner control are evaluated separately [8], [9], [26].

The experiment uses simulated bookings and weather, with known booking and cancellation mechanisms. No customer records or external weather observations were used. The hidden generator parameters remain separate from model inputs and support checks after fitting. Real-data validation would require decision-time weather records, a data-sharing agreement and any necessary privacy and institutional review. Open-Meteo and ERA5 are possible external weather sources for that later stage [36], [37].

B. Simulation data and preprocessing

The generator creates 14 venues, 730 days and 15 hourly slots per venue-day, from 08:00 through 22:00. Facility types are futsal, football, tennis and swimming. Each row represents an offered slot, including unbooked capacity. The binary target is whether a booking occurred; cancellations are stored separately and do not change that target. Venue identifiers are synthetic, and the dataset contains no customer identifiers.

Booking probability is generated from a logistic mechanism with facility type, venue popularity, peak hours, weekday, simulated weather, calendar indicators, lead time and relative price. The price rule includes predictable peak and weekend components and random variation. The simulator sets different price-response coefficients by facility type and peak status. Cancellation probability depends on relative price and lead time. Weather fields are noisy simulated forecasts, and holiday and examination flags are synthetic indicators rather than verified observations from official calendars. The seed is 2026. The hidden booking intercept and price coefficient are written to a separate file and used only after fitting to check the simulated counterfactuals.

Cleaning removes duplicate venue–time keys and nonpositive or missing prices. The code reports both the number flagged and the number actually removed before constructing the analysis splits. Hour, weekday and month are encoded using sine and cosine terms. Lead times are grouped into less than 3 hours, 3–24 hours, 1–7 days and more than 7 days. Relative price is $r = p/p_{\mathrm{ref}}$, where p_{ref} is the venue reference price; its logarithm is the price feature. Forecasting uses weather, calendar, lead time, venue type and venue identity. Category vocabularies and scaling parameters are learned from training data.

The logistic, LightGBM and LSTM candidates receive the same covariates for the current slot and the previous three offered slots at that venue. The first observations are padded with current-slot covariates. These windows contain no booking outcomes or cancellation labels. Flattened windows are used by logistic regression and LightGBM; the LSTM receives the corresponding four-step sequence. This is a short contextual sequence experiment, not a test of long-range temporal memory. The simulation assumes that these published-slot covariates are available at decision time. The dataset contains no forecast issue timestamps to verify that assumption on real data.

C. Temporal validation and candidate models

The final 20% of the timeline is held out for testing. Four expanding-window folds partition the earlier period, with a seven-day gap between each fold's training endpoint and validation start. The final models use all pre-test observations. No additional seven-day gap is imposed at the final test boundary; the feature windows contain no lagged outcomes. This distinction is retained in the recorded split metadata. Chronological validation follows time-series evaluation principles [14].

Logistic regression uses standardized covariates and an L_2 penalty. LightGBM uses a non-increasing constraint on the current log-price feature [38]. The LSTM has one recurrent layer and a sigmoid probability output, trained with binary cross-entropy and Adam. Both learned nonlinear models are repeated with seeds 0–4. Logistic regression is deterministic for this implementation and is fitted once after tuning. Model selection minimizes mean validation log-loss, before test metrics are computed. Final probabilistic comparisons use the average prediction across the five seeds; individual-seed metrics are also retained.

Optuna's tree-structured Parzen estimator (TPE) sampler uses seed 2026 [39]. The pilot budget is eight trials per candidate, each evaluated on all four folds. Logistic regression tunes regularization strength. LightGBM tunes tree count, learning rate, leaf count, minimum leaf size and regularization. The LSTM tunes hidden width, learning rate, weight decay and a fixed budget of five or eight epochs. The search uses a fixed training budget on the available central processing unit (CPU). Equal trial counts do not imply equal computation. No Temporal Fusion Transformer is fitted; the recurrent model is the neural alternative in this pilot [40].

Two additional baselines are evaluated. Climatology is the training booking rate for each venue, hour and weekday, with the overall training mean as a fallback. The seasonal autoregressive integrated moving average model with exogenous regressors (SARIMAX) uses separate hourly occupancy sequences for each venue. Its regressors include weather, calendar, lead time and price. A daily seasonal autoregressive term uses the 15 offered slots per day. The training Akaike information criterion (AIC) selects between nonseasonal orders $(0,0,0)$ and $(1,0,0)$. The fitting limit is 60 optimizer iterations, and convergence flags are retained. Forecasts cover the complete test horizon without assimilating test labels, and are clipped to $[10^{-6}, 1-10^{-6}]$ for probability scoring. SARIMAX is an AIC-selected benchmark, rather than a candidate selected by the same validation log-loss search.

D. Price response and counterfactual demand

Historical prices can be correlated with the demand conditions used to choose them. The price-response step estimates the partially linear relation

$$y = \theta(z)\ln r + g(x) + \epsilon, \qquad \ln r = m(x) + v. \tag{1}$$

The segment z combines facility type, lead-time bin and peak status. LightGBM nuisance regressors estimate g and m using past-only expanding folds with the same seven-day gaps. The initial block supplies training history and contributes no cross-fitted residuals. Segment coefficients are estimated from residualized outcomes and prices, with venue-week clustered standard errors. This uses the orthogonalization idea from double/debiased machine learning (DML) [16]. A segment is eligible for pricing when it has at least 500 residual observations and a residual-to-observed log-price variance ratio of at least 0.05.

The response exponent is $\widehat{\varepsilon}(z) = \widehat{\theta}(z)/\overline{y}(z)$. Because the generator's probability function is logistic, this constant-exponent layer is an approximation. We compare its coefficients with the simulator's average local probability derivative as a diagnostic. This comparison is not an exact coverage test for the partially linear projection parameter. Naive coefficients from unadjusted regressions are recorded on the same residual-estimation sample.

The selected price-aware predictor is evaluated with current relative price set to one, while other covariates are held fixed, to obtain $\widehat{D}_0(x)$. Demand at an alternative price is

$$\widehat{D}(p,x) = \min\{1, s\widehat{D}_0(x)(p/p_{\mathrm{ref}})^{\widehat{\varepsilon}(z)}\}. \tag{2}$$

We choose the scalar s from training observations only. It makes modeled revenue at historical prices equal observed training revenue. Test-period agreement is reported separately. Setting current price explicitly avoids treating a price-omitting predictor as a reference-price counterfactual.

E. Optimization and operational outcomes

Cancellation probability $q(p,x)$ is estimated with a separate LightGBM classifier on booked training slots. The objective is

$$p^* = \arg\max_{p\in\mathcal{P}} \{p - c - \phi q(p,x)\}\widehat{D}(p,x). \tag{3}$$

For this simulation, $c = 0.30p_{\mathrm{ref}}$, $\phi = 0.50p_{\mathrm{ref}}$, and owner bounds are $0.70p_{\mathrm{ref}}$ to $1.30p_{\mathrm{ref}}$. These cost and penalty fractions are fixed simulation assumptions. A grid uses steps of $0.01p_{\mathrm{ref}}$ within both the owner bounds and each segment's training 2.5th–97.5th percentile price support. The historical price is included as a no-change option, even if it falls outside the trimmed support. Cancellation probability is recomputed at every candidate price. Ineligible segments retain their historical price.

Revenue is gross booking revenue, $R(\pi) = \sum_t \pi_t \widehat{D}(\pi_t, x_t)$, and its relative change is

$$\Delta R = 100\{R(p^*)/R(p_0) - 1\}. \tag{4}$$

Both policies are evaluated under the same demand model. Revenue is distinct from the cancellation-adjusted profit used for optimization. Utilization is expected booked slots divided by offered slots. Off-peak occupancy uses weekdays from 10:00 up to, but not including, 16:00. The cancellation rate is expected cancellations divided by expected bookings. Changes in utilization and cancellation rates are expressed in percentage points. This fixed window is not assumed to have the lowest occupancy.

F. Metrics, uncertainty and success criteria

The main probability scores are Brier score and log-loss [11], [12]:

$$\text{Brier} = N^{-1} \sum_t (y_t - \widehat{p}_t)^2, \tag{5}$$

$$\text{LogLoss} = -N^{-1} \sum_t [y_t \ln \widehat{p}_t + (1 - y_t) \ln(1 - \widehat{p}_t)]. \tag{6}$$

Additional measures include area under the receiver operating characteristic curve (ROC-AUC) and expected calibration error (ECE) in ten equal-width bins. We also report calibration slope, calibration intercept and Brier skill score (BSS) relative to climatology. Calibration is assessed without using test labels to alter predictions [13]. We sum predicted probabilities by venue-day to calculate mean absolute error (MAE) in bookings per day. We also report weighted absolute percentage error (WAPE). We omit slot-level mean absolute percentage error (MAPE) because the target includes zeros [15].

Forecast intervals use 2,000 venue-week block bootstrap replicates. Paired loss differences use the same resampled blocks. Two-sided bootstrap tests center the bootstrap distribution at its observed difference and apply a finite-replicate correction. Holm correction covers the eight primary comparisons against the selected model. Diebold and Mariano provide a related framework for comparing predictive accuracy; we use the paired block bootstrap described here [41]. The selected model's calibration slope uses 300 block replicates. Revenue and operational intervals use 1,000 bootstrap replicates combined with segment-level normal draws of the estimated elasticities. Only blocks present in the test data are sampled. The fitted predictors and chosen policy remain fixed, so these intervals do not propagate all model-fitting or policy-selection uncertainty. Venue-week resampling also treats those blocks as independent despite the simulator's shared daily temperature component; this assumption can understate uncertainty from common weather shocks.

H1 requires lower Brier score and log-loss against every evaluated baseline at adjusted $p < 0.05$, positive Brier skill, and a calibration-slope interval containing one. H2 requires a revenue interval with lower bound above zero; no agreed commercial implementation threshold is available. H3 requires a positive lower bound for off-peak occupancy and a cancellation-change upper bound below half the training weekly cancellation-rate standard deviation. These criteria are evaluated on the simulation. There is no observed paired counterfactual revenue outcome from which to estimate a minimum detectable commercial effect.

G. Additional analyses and computation

Additional analyses compare estimated price effects with simulator derivatives and evaluate the chosen prices under the full booking and cancellation functions. They also vary elasticity by a common factor from 0.5 to 1.5. Other checks remove weather from LightGBM and report errors by facility type and test season. The weather ablation reuses the selected LightGBM settings and all five seeds; it does not retune after inspecting test results.

The generator, tuning and forecast bootstrap use seed 2026. Model seeds are 0–4, and the combined elasticity and policy bootstrap uses seed 123. Models use two computation threads on a platform that exposes nine logical CPUs. Saved artifacts record split boundaries, search trials, selected settings, per-seed scores, predictions, paired tests and policy diagnostics. The verification script checks test membership, temporal gaps, metrics, the price grid and revenue arithmetic. Section VII gives the environment and execution commands.

IV. Results

A. Study status and research questions

RQ1 assesses booking-probability forecasts, RQ2 assesses revenue from constrained prices, and RQ3 assesses off-peak occupancy and cancellations. Each question uses the H1–H3 criteria defined in Section III. Models and policies share the final test period. Hidden generator parameters were excluded from training, selection and optimization; they were used only after fitting to check the final policy.

B. Data and test period

Table I records the final analysis data. The 153,300 rows cover 730 days and 14 venues, with 15 offered slots per venue-day. The first 584 days form the development period, and the last 146 days form the test period. There are 31,888 booked and 90,752 unbooked development slots, compared with 9,988 booked and 20,672 unbooked test slots. Booking prevalence is 26.00% and 32.58%, respectively. The development period contained 3,187 cancellations among booked slots, and the test period contained 958. These correspond to 9.99% and 9.59% of bookings. Cancelled bookings remain positive examples for the booking target; the cancellation classifier models their later cancellation separately. No cancelled rows were discarded from the forecasting comparison. All 15 dataset columns are complete. The audit found no duplicate venue–time keys, nonpositive prices or missing prices, so all records were retained.

The generator reproduced the archived comma-separated values (CSV) file byte for byte with seed 2026. The same generation run recovered the hidden booking intercepts and price coefficients used in the simulator checks. Training windows expand across the development folds and therefore overlap. Adding their training and validation counts would count some observations more than once. Each validation fold is separated from its training observations by seven days. The held-out test observations occur from 8 April to 31 August 2026 and contain spring and summer dates only.

C. RQ1 Booking probability forecasts

Mean validation log-loss was 0.5031 for LightGBM, 0.5137 for the LSTM and 0.5329 for logistic regression. LightGBM was selected before evaluating the held-out test period. Table II reports the common test comparison. The selected model's Brier score was 0.1951, compared with 0.2107 for climatology, 0.2092 for logistic regression, 0.1954 for the LSTM and

TABLE I
FINAL SYNTHETIC DATASET AND CHRONOLOGICAL VALIDATION SIZES. DEVELOPMENT INCLUDES ALL PRE-TEST OBSERVATIONS; FOLD WINDOWS OVERLAP. ZERO ROWS WERE REMOVED.

Item	Value
Source	Synthetic booking simulator
Period	2024-09-01 to 2026-08-31
Venues / facility types	14 / 4
Rows before / after cleaning	153,300 / 153,300
Duplicates / invalid prices / missing cells	0 / 0 / 0
Development / test rows	122,640 / 30,660
Booked / unbooked development rows	31,888 / 90,752
Booked / unbooked test rows	9,988 / 20,672
Booking prevalence development / test	26.00% / 32.58%
Fold 1 train / validation	23,100 / 24,514
Fold 2 train / validation	47,614 / 24,472
Fold 3 train / validation	72,086 / 24,514
Fold 4 train / validation	96,600 / 24,570

0.2038 for SARIMAX. Its log-loss was 0.5747. All four fitted models had positive Brier skill relative to climatology.

The selected LightGBM configuration used 240 trees, a learning rate of 0.06824, seven leaves per tree and a minimum of 80 observations per leaf. The selected LSTM had 24 hidden units and eight training epochs, with a learning rate of 0.00597. Logistic regression used $C = 0.00103$. The validation search selected these configurations before their performance on the test set was known.

For LightGBM and the LSTM, the main comparison uses predictions averaged across the five model seeds. The individual LSTM runs had mean Brier score 0.1962, with standard deviation 0.0003. Their mean log-loss was 0.5772, with standard deviation 0.0006. LightGBM's fixed, deterministic configuration produced identical probability scores across the five recorded seeds. Sampling uncertainty is reported separately through the intervals in Table II. Logistic regression was fitted once after tuning, and SARIMAX contributed one AIC-selected fit per venue.

The paired comparisons in Table III report the loss differences and the evidence from the paired statistical tests. LightGBM had lower Brier score and log-loss than climatology, logistic regression and SARIMAX, with Holm-adjusted $p = 0.0040$ for each comparison after rounding. The differences against the LSTM were much smaller: -0.0003 in Brier score and -0.0006 in log-loss. Both paired intervals included zero, and their adjusted p-values were 0.1859 and 0.2369. Because the comparison with the LSTM did not establish lower loss, H1 was not satisfied.

The selected model's calibration slope was 1.0385, with a 95% interval of 1.0019 to 1.0768; its intercept was 0.0393. The slope interval excluded one, so the calibration component of H1 also failed. Figure 1 plots predicted and observed booking probabilities. The LSTM had ECE 0.0038, compared with 0.0043 for LightGBM. LightGBM remained the model selected by validation log-loss; test calibration was reported without changing that choice. All 14 selected venue-level SARIMAX fits reported optimizer convergence.

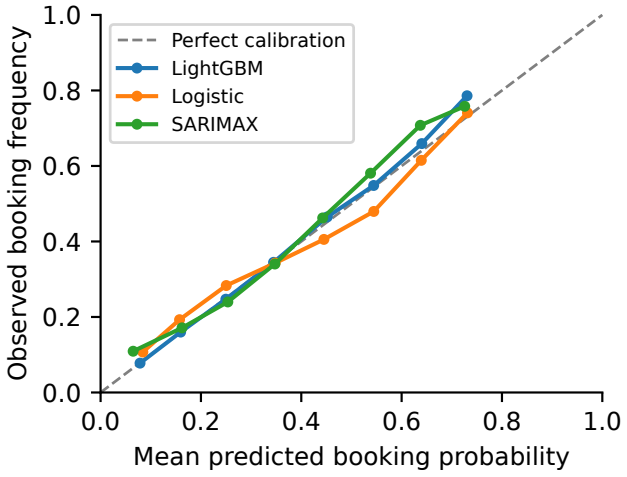


Fig. 1. Test-set reliability on synthetic slots. Both axes are dimensionless probabilities. Points show ten equal-width probability bins with at least 20 observations; the dashed diagonal indicates equal predicted and observed booking rates.

At the venue-day level, the LSTM had MAE 1.371 bookings per day and WAPE 28.05%, compared with 1.378 and 28.19% for LightGBM. Table V gives the remaining values. These aggregated measures were not assessed with a separate paired test, so the reported ordering is descriptive. Aggregation changes the scored quantity from individual booking probabilities to the expected number of bookings across each venue-day.

D. RQ2 Price response and counterfactual revenue

Cross-fitting supplied 98,070 residual observations across 32 segments. Segment counts ranged from 531 to 7,195. The residual-to-observed log-price variance ratio ranged from 0.812 to 0.867, and all 32 segments met the sample-size and residual-variance thresholds. Five estimated probability derivatives were positive, although the simulator's local booking response to price was negative. Figure 2 shows the estimates and their intervals in the segment order retained in the accompanying CSV.

Against the simulator's average local derivative, the mean absolute discrepancy was 0.1647 for the orthogonalized estimate and 0.3125 for the naive estimate. The simulator derivative lay inside 29 of the 32 estimated intervals, or 90.6%. This check compares the fitted partially linear coefficient with the average local derivative of the nonlinear generator. The two quantities have different statistical targets, so interval overlap is reported as a diagnostic.

The training revenue normalization factor was 1.0341. On test data, modeled revenue at historical prices divided by observed booking revenue was 0.9882. The corresponding modeled booking rate was 32.38%, compared with the observed 32.58%. The policy changed prices for 97.57% of slots, and the mean recommended-to-historical price ratio was 1.0358. There were 20,259 price increases, 9,655 decreases and 746 unchanged prices, counting differences smaller than

TABLE II
MAIN TEST RESULTS ON 30,660 SYNTHETIC SLOTS. BRACKETS GIVE 95% VENUE-WEEK BOOTSTRAP INTERVALS. ALL VALUES ARE DIMENSIONLESS. LOWER BRIER, LOG-LOSS AND ECE ARE PREFERABLE; LARGER AUC AND BSS ARE PREFERABLE. CALIBRATION TARGETS ARE SLOPE 1 AND INTERCEPT 0.

Model	Brier [95% CI]	Log-loss [95% CI]	AUC	ECE	BSS	Slope	Intercept
Climatology	0.2107 [0.2058, 0.2155]	0.6181 [0.6062, 0.6301]	0.6453	0.0665	0.0000	0.6912	0.0388
Logistic	0.2092 [0.2053, 0.2128]	0.6072 [0.5983, 0.6156]	0.6355	0.0301	0.0071	0.7486	-0.1628
LightGBM	0.1951 [0.1915, 0.1984]	0.5747 [0.5665, 0.5823]	0.6973	0.0043	0.0740	1.0385	0.0393
LSTM	0.1954 [0.1919, 0.1987]	0.5753 [0.5671, 0.5830]	0.6964	0.0038	0.0724	1.0010	0.0051
SARIMAX	0.2038 [0.2003, 0.2071]	0.5957 [0.5878, 0.6032]	0.6612	0.0152	0.0326	1.0516	0.0472

TABLE III
PAIRED LOSS DIFFERENCES, LIGHTGBM MINUS THE NAMED BASELINE. NEGATIVE VALUES FAVOUR LIGHTGBM. INTERVALS USE 2,000 VENUE-WEEK REPLICATES; ADJUSTED P-VALUES APPLY HOLM CORRECTION ACROSS EIGHT TESTS.

Baseline	Loss	Difference	95% CI	Adjusted p
Climatology	Brier	-0.0156	[-0.0180, -0.0132]	0.0040
Climatology	Log-loss	-0.0434	[-0.0500, -0.0370]	0.0040
Logistic	Brier	-0.0141	[-0.0154, -0.0129]	0.0040
Logistic	Log-loss	-0.0325	[-0.0355, -0.0296]	0.0040
LSTM	Brier	-0.0003	[-0.0007, 0.0001]	0.1859
LSTM	Log-loss	-0.0006	[-0.0015, 0.0004]	0.2369
SARIMAX	Brier	-0.0087	[-0.0097, -0.0078]	0.0040
SARIMAX	Log-loss	-0.0210	[-0.0235, -0.0188]	0.0040

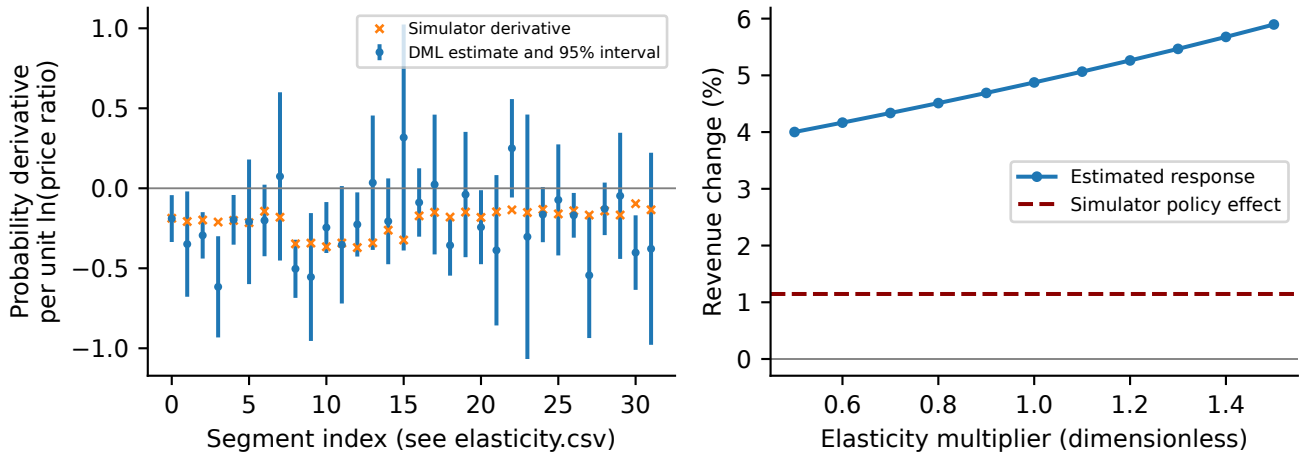


Fig. 2. Price-response diagnostics on synthetic data. Left: estimated segment probability derivatives with 95% clustered intervals and simulator average local derivatives; the segment index follows the saved elasticity table. Right: revenue change for the fixed chosen policy when estimated elasticities are multiplied by 0.5–1.5. The horizontal dashed line gives the known simulator effect for that policy.

one millionth of a currency unit as unchanged. The no-change option remained available in the optimization. The demand diagnostics use the fitted price-response layer; the next comparison evaluates the same policy with the known generator.

Table IV reports a modeled revenue change of 4.87%, with a 95% interval of 3.28 to 6.79%. Its lower bound exceeded the zero threshold, so the estimated-demand calculation satisfied H2. Evaluating exactly the same recommended prices under the simulator's known booking function gave 1.15%. That value was below the lower endpoint of the estimated interval. The absolute difference between estimated and simulator revenue changes was 3.73 percentage points.

With the chosen policy held fixed, multiplying all estimated elasticities by a common factor from 0.5 to 1.5 produced revenue changes from 4.00% to 5.90%. Figure 2 shows the

TABLE IV
RECOMMENDED MINUS HISTORICAL POLICY ON THE SAME SYNTHETIC TEST SLOTS. ESTIMATED INTERVALS COMBINE VENUE-WEEK RESAMPLING AND 1,000 ELASTICITY DRAWS. THE SIMULATOR COLUMN EVALUATES THE FIXED CHOSEN PRICES USING THE KNOWN MECHANISM AND IS A DETERMINISTIC EXPECTATION, NOT OBSERVED CUSTOMER REVENUE.

Outcome	Estimate	95% CI	Simulator
Revenue change (%)	4.87	[3.28, 6.79]	1.15
Utilization change (pp)	0.56	[0.02, 1.16]	-0.56
Off-peak change (pp)	-0.31	[-0.94, 0.33]	-0.92
Cancellation change (pp)	-0.51	[-0.69, -0.32]	-0.94

complete sensitivity curve. The calculated changes remained positive across the tested factors; no break-even factor was observed in this interval. The horizontal simulator value evaluates the same policy under its full known response, whereas the curve changes one shared multiplier in the estimated response model.

E. RQ3 Off-peak occupancy and cancellations

Under estimated demand, overall utilization changed by 0.56 percentage points, with an interval of 0.02 to 1.16. The simulator evaluation gave -0.56 percentage points for the same prices. The estimated response predicted an increase, while the simulator response gave a decrease in overall utilization. Off-peak occupancy changed by -0.31 percentage points under estimated demand, with an interval of -0.94 to 0.33; the corresponding simulator change was -0.92 percentage points.

The estimated cancellation-rate change was -0.51 percentage points, with an interval of -0.69 to -0.32. The training-derived non-inferiority margin was 0.75 percentage points, and the estimated upper bound was below that margin. The cancellation condition passed, but the lower bound for off-peak occupancy did not exceed zero. H3 was therefore not satisfied. Both cancellation rates use expected bookings as the denominator. Offered slots without a booking are excluded from that denominator.

F. Additional analyses and observed errors

Removing the four weather variables from every covariate step increased LightGBM's Brier score from 0.1951 to 0.1974 and log-loss from 0.5747 to 0.5806. The paired Brier difference was 0.0023, with a 95% interval of 0.0017 to 0.0029. The corresponding log-loss interval was 0.0044 to 0.0073. The ablation was evaluated separately from the eight primary tests. Its bootstrap intervals are descriptive, and no multiplicity-adjusted significance is claimed for these additional comparisons.

Figure 3 compares observed and predicted booking rates by facility type. Football had an observed rate of 34.25% and a predicted rate of 32.97%, the largest absolute rate discrepancy among the four types. Swimming had the highest group Brier score, 0.2091; tennis had the lowest, 0.1845. Booking prevalence differs across these groups. The scores describe each group's prediction errors without establishing a controlled comparison of facility difficulty. The saved group table includes denominators, booked counts and log-loss for every type.

A further descriptive error check examined the 1,875 test slots with predicted booking probability of at least 0.60. Their mean prediction was 65.74%, and 1,280 were booked, an observed rate of 68.27%. The remaining 595 slots were unbooked despite receiving predictions above that threshold. The 0.60 threshold was used for this descriptive error check only. It had no role in model selection or recalibration.

The seasonal breakdown contains 11,340 spring slots and 19,320 summer slots. LightGBM's Brier score was 0.1793 in spring and 0.2043 in summer; the observed booking rates were 27.72% and 35.43%. No autumn or winter test scores were available. The test period therefore provides two seasonal groups, while all four seasons occur in the development data.

TABLE V
VENUE-DAY ERRORS AFTER SUMMING SLOT PROBABILITIES. MAE IS MEASURED IN BOOKINGS PER VENUE-DAY AND WAPE IS A PERCENTAGE. THESE SECONDARY MEASURES USE THE SAME TEST SLOTS AS THE MAIN PROBABILITY COMPARISON.

Model	MAE (bookings/day)	WAPE (%)
Climatology	1.810	37.04
Logistic	1.677	34.32
LightGBM	1.378	28.19
LSTM	1.371	28.05
SARIMAX	1.400	28.64

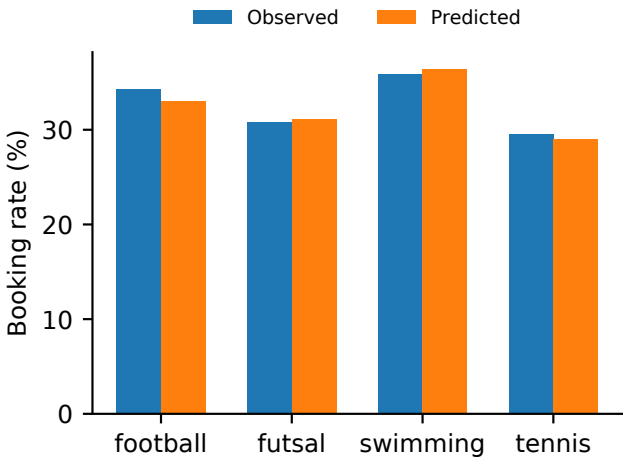


Fig. 3. Observed and LightGBM-predicted booking rates by synthetic facility type on the common test period. Football and futsal each have 8,760 slots; swimming and tennis each have 6,570. Bars show rates as percentages of offered slots.

Several parts of the pipeline did not meet the specified checks. LightGBM's loss advantage over the LSTM was not supported, five estimated price derivatives were positive, and the simulator revenue effect fell outside the estimated interval. Estimated and simulator utilization changes also had opposite signs. H1 and H3 did not satisfy all their conditions. The selected model, price policy, test set and success thresholds were kept fixed after these outcomes were observed. Section V states their implications and remaining limits.

V. CONCLUSION

This study evaluated a forecasting and constrained-pricing pipeline on synthetic sports slots. The common evaluation record made it possible to check booking predictions, estimated price responses and policy outcomes on the same inventory. The findings show why a forecast comparison alone cannot validate the revenue gains reported by a price optimizer.

For RQ1, LightGBM improved on historical booking rates, logistic regression and the seasonal statistical baseline in both probability-loss comparisons. The paired tests did not establish an advantage over the LSTM. Its calibration-slope interval also excluded one. H1 therefore failed, despite positive Brier skill. These findings concern the selected configurations and

training budget, and do not support a general ranking of model architectures.

For RQ2, the estimated response model gave a positive revenue lower bound and passed H2. The same chosen prices produced a smaller expected gain under the known simulator mechanism, outside the estimated interval. A favorable offline estimate therefore overstated the policy effect even in a setting where the full booking mechanism was available for an independent check. The uncertainty calculation did not capture this response-model error.

For RQ3, the cancellation-change upper bound met the non-inferiority condition, but off-peak occupancy did not show an increase. H3 failed because it required both conditions. The simulator check also showed lower off-peak use. A policy that improves estimated revenue cannot therefore be described as improving every operational outcome.

The practical value is the ability to trace these disagreements through the pipeline. Pricing reports should distinguish model-estimated revenue, expectations under a known simulator and revenue observed after an intervention. They should also separate gross booking revenue from the cancellation-adjusted profit used by the optimizer. Reporting both makes the decision objective and its consequences easier to assess.

The choice of denominator must also stay visible. Forecasting uses every offered slot, while cancellation rates refer to expected bookings. A lower cancellation rate does not show that more quiet-hour capacity was used. Operators need these outcomes side by side to judge a price policy against their service goals. Keeping the goals separate also prevents a positive revenue estimate from being treated as evidence for improved off-peak use.

The main limits are one generator, restricted tuning, short recurrent sequences and a test period limited to spring and summer. The partially linear price-response approximation differs from the nonlinear generator; five estimated derivatives had the wrong sign. The bootstrap conditions on the fitted models and chosen policy. It can also miss uncertainty from shared daily weather shocks. These limits restrict the interpretation of the intervals and transfer to other settings.

Further work should test a better-matched price-response function, larger training budgets and leave-venue-out evaluation. Resampling should account for common time shocks and model-selection uncertainty. A real-data study needs complete offered-slot inventory, decision-time covariates and the required privacy review. A prospective price experiment could then test revenue and off-peak outcomes under owner-defined limits.

VI. Data availability

The dataset *Sports Bookings Synthetic* contains 153,300 generated venue-time slots from 14 synthetic venues and no customer identifiers. It is available in the Zenodo repository at https://doi.org/10.5281/zenodo.24080033 (release v1.0, commit 424d314 of https://github.com/Asslan77/assignment_3, directory `sports_pricing_assignment/data/`) under a CC BY 4.0 licence. The file `sports_bookings_synthetic.csv` contains the analysis data. The same directory provides the generation settings (seed 2026) and the hidden simulator parameters used only for checks after model fitting, and `MANIFEST.sha256` records file digests for verification. No real aggregator records or downloaded external weather observations were used.

VII. Code availability

All code used for data generation, model training, price optimization and evaluation is available at https://github.com/Asslan77/assignment_3 (release v1.0, commit 424d314, DOI: 10.5281/zenodo.24080033), in the directory `sports_pricing_assignment/`, under the MIT licence. The recorded run used Python 3.12.14 on Linux with an AMD EPYC 9V74 CPU and no graphics processing unit (GPU). Library versions were NumPy 2.3.5, pandas 2.2.3, scikit-learn 1.8.0 and SciPy 1.17.0. Models used LightGBM 4.7.0, Optuna 5.0.0, PyTorch 2.14.1+cpu and statsmodels 0.15.0. The exact environment is specified in `requirements.txt` and `environment.json`. From the `sports_pricing_assignment/` directory, `python src/run_analysis.py` runs the experiment and `python src/check_artifacts.py` verifies the saved outputs. A clean analysis run took about six minutes on a CPU, excluding package installation and manuscript compilation. The repository also includes a notebook, numerical tables and figure scripts. No proprietary software was used, and third-party dependencies retain their separate licences.

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